

Database Management Systems

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Project Scope:

Gaming Store Management System is designed to connect gamers with individual game developers and major publishers, creating a diverse platform that caters to the needs of both creators and players. Our primary focus is to help indie developers and users not to worry about bot reviews, malicious software, or hidden charges. We provide a stage for showing fresh, creative ideas and empowering independent creators to reach an audience.

Key Features and Objectives:

- Support for Indie Developers
- Scam-Free Ecosystem
- Authentic Reviews
- User-Friendly Design
- Community Engagement
- Transparency

Requirements:

Customer information:

- Name, Age, Location
- Preferences
- Reviews given
- In-game achievements

Publisher information:

- Name, Service number, Licenses (anything legal)
- Blogs/Newsletters/Updates
- Verifications done & terms of services
- Average rating received

Payment Portal Information:

- Name, payment methods (Credit/Debit, UPI)
- PCI DSS compliance, Fraud detection
- Setup and maintenance, transaction fees
- compatibility, Latency & transaction speed

Features:

1. User onboarding

1.1 Publishers

- Registration of a new account
- Listing their games
- Analytics and revenue

1.2 Customers

- Registration of a new account
- Selecting payment methods
- Billing account details

2. Store

2.1 Search

- Genre-based categorization
- Searching for a title
- Filtering and sorting based on ratings

2.2 Review

- Authenticate as a genuine user
- Write a review
- Report a review

2.3 Sales

- Clearance sales on old titles
- Discounts on most wish-listed games
- Free giveaway for some number of users

3. Forum

- Publishers can generate hype for upcoming releases
- New gameplay/tricks ideas by users
- FAQs and posts open to address technical/(non) issues
- Blogposts to share game arts or ideas

4. Order management

- Cart to add/keep games
- Order history to view past purchases
- Check the total and proceed to checkout

Entities:

1. Customer

1.1 Attributes

- *Customer_id - Unique identifier for each customer
- Name - Full name of the customer
- Billing email - Email address used for billing and communication.
- Payment methods - Preferred payment options are saved by the customer.

1.2 Relationships

- *Customer* BUYS *Games*
- *Customer* REVIEWS/POSTS *Games* on *Forum*
- *Customer* ADDS *Games* to *Order*

2. Publisher

2.1 Attributes

- *Publisher_id - Unique identifier for each publisher.
- *License_number - Legal license number for game publishing.
- Type(company/individual) - Specifies if the publisher is a company or an individual developer.

2.2 Relationships

- *Publisher* **LISTS** *Game*
- *Publisher* **POSTS** on the *Forum*
- *Publisher* **ADDRESSES** *Moderator*

3. Forum

3.1 Attributes

- *Forum_ID - Unique identifier for each forum discussion.
- Title - Name or subject of the forum thread.
- Type(Customer/publisher) - Specifies whether the forum post is created by a customer or a publisher.
- Description - Details on the topic of discussion.
- Analytics - Post impressions/stats.

3.2 Relationships

- *Forum* **CONTAINS** *Blogs*
- *Forum* **HAS** **REVIEWS** from *Customers*.
- *Forum* is **MODERATED** by *Moderators*.

4. Order

4.1 Attributes

- *Order_id– Unique identifier for each order.
- Customer_id – Identifier linking the order to a specific customer.
- Payment_method – The payment option used for the order (Credit/Debit, UPI, etc.).
- Billing_email - Email address associated with the order's billing details.
- Total - The total amount paid for the order.

4.2 Relationships

- *Order* **PLACED BY** *Customer*
- *Order* **CONTAINS** *Games*

5. Game

5.1 Attributes

- *Game_ID - Unique identifier for each game.
- Publisher_ID - Identifier linking the game to its publisher.
- Price - Cost of the game for purchase.
- Genre - Category or type of the game (e.g., Action, RPG, Strategy).
- Title - Name of the game.
- Rating - Average user rating based on reviews.
- Analytics - Unique / Total no. of downloads.

5.2 Relationships

- *Game* PURCHASED by *Customer*
- *Game* PUBLISHED by *Publisher*
- *Game* can be POSTED on *Forum* by *Publisher*
- *Game* HAS a *Blog*

6. Cart

6.1 Attributes -

- *Customer_ID – Identifier linking the cart to a specific customer.
- Total - The total cost of all the items currently in the cart.
- Items - List of items (games) added to the cart.

6.2 Relationships

- *Cart* CONTAINS *Order*
- *Cart* ACCESSED by *Customer*

7. Moderator

7.1 Attributes

- *Mod_id - Unique identifier for each moderator.
- Genre - Area of expertise or focus of the moderator (e.g., RPG, Action).
- Issues_resolved - Number of issues or complaints successfully resolved by the moderator.
- Engagement – Level of interaction or activity of the moderator with the community (e.g., posts, responses).

7.2 Relationships

- *Moderator MANAGES Forum.*

8. Blog(Weak entity)

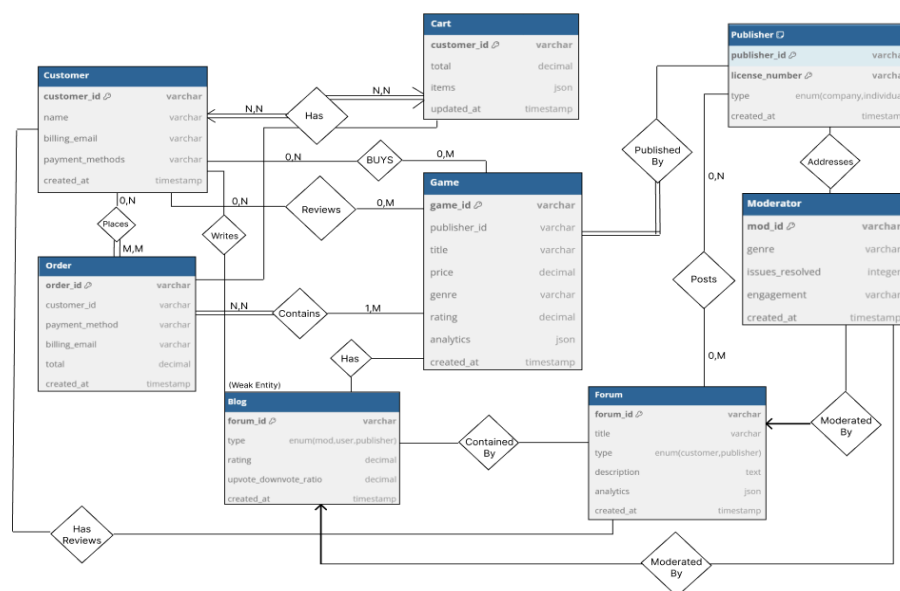
8.1 Attributes

- *Forum_id - Unique identifier for each forum post.
- Type(mod/user/publisher post) - Specifies whether the post is made by a moderator, user, or publisher.
- Rating - Average rating given to the forum post by users.
- Upvote-downvote ratio - Ratio of upvotes to downvotes received on the forum post.

8.2 Relationships

- *Blog* is WRITTEN BY *Customer/Publishers/Moderators*.
- *Blog* is MODERATED by *Moderators*.

ER diagram:



[Figma link](#) for the ER diagram

